MERN Stack Viva Preparation – TY B.Sc. IT

I'll prepare you for your external practical viva using your uploaded 16-practical MERN guide.

The goal is to help you answer confidently even if the professor asks questions directly from your code.

We'll cover:

1. Basic MERN Stack and technology concepts.
2. Viva questions and answers for all 16 practicals.
3. Important code explanations.
4. Common external viva questions and tricky follow-up questions.
5. A mock viva to test your preparation.

Part 1: Basic MERN Stack viva

1. What is MERN Stack?

MERN is a full-stack JavaScript technology used to develop web applications.

- M – MongoDB: Database used to store data.
- E – Express.js: Backend web framework.
- R – React.js: Frontend JavaScript library.
- N – Node.js: JavaScript runtime environment.

2. What is full-stack development?

Full-stack development means developing both the frontend and backend of a web application, including database management.

3. What is JavaScript?

JavaScript is a programming language used to make web pages interactive. It can also be used on the server side with Node.js.

4. What is Node.js?

Node.js is a JavaScript runtime environment that allows JavaScript code to run outside the browser.

5. What is Express.js?

Express.js is a lightweight web framework for Node.js used to create servers, routes and APIs.

6. What is React.js?

React is a JavaScript library used to build interactive user interfaces using reusable components.

7. What is MongoDB?

MongoDB is a NoSQL database that stores data in document format, similar to JSON.

8. What is the difference between SQL and NoSQL?

SQL	NoSQL
Relational database	Non-relational database
Stores data in tables	MongoDB stores data in documents
Uses rows and columns	Uses documents and collections
Example: MySQL	Example: MongoDB

9. What is an API?

API stands for Application Programming Interface. It allows two applications to communicate and exchange data.

10. What is npm?

npm stands for Node Package Manager. It is used to install and manage JavaScript packages.

11. What is a package?

A package is reusable code or a library that can be installed and used in a project.

12. What is the difference between frontend and backend?

- Frontend: The part of a website that users see and interact with.
- Backend: The server-side part that handles logic, requests and database operations.

Part 2: Practical-wise viva questions and answers

These questions follow the exact practicals in your uploaded guide. Learn the bold terms because examiners often ask follow-up questions about them.

Practical 1 – JavaScript operators and control statements

Q1. What are operators in JavaScript?

Operators are symbols used to perform operations on variables and values.

Q2. Name different types of operators.

- Arithmetic: +, -, *, /, %
- Comparison: >, <, ==, ===
- Logical: &&, ||, !
- Assignment: =, +=, -=

Q3. What is a control statement?

A control statement controls the flow of program execution.

Q4. What is the difference between if-else and for loop?

- if-else is used for decision-making.
- for loop is used to repeat statements.

Q5. What is the use of % operator?

It returns the remainder after division.

Example: 12 % 5 = 2.

Practical 2 – Display a message using server response

Q1. What is a server?

A server is a computer or software that receives client requests and sends responses.

Q2. What is Express?

Express is a Node.js framework used to create web servers and APIs.

Q3. What is require("express")?

It imports the Express module into the JavaScript file.

Q4. What is app.get()?

It defines a route that handles HTTP GET requests.

Q5. What is res.send()?

It sends a response from the server to the client.

Q6. What is localhost?

Localhost refers to your own computer, usually represented by IP address 127.0.0.1.

Practical 3 – Print even numbers from 1 to 50

Q1. What is a loop?

A loop repeatedly executes a block of code until a condition becomes false.

Q2. Which loop is used in this practical?

The for loop.

Q3. How do you identify an even number?

An even number is divisible by 2 and gives remainder 0.

Q4. Explain this condition:

if (number % 2 === 0)

It checks whether the number is even.

Q5. What is number++?

It increases the value of number by 1.

Practical 4 – Different routes in Express

Q1. What is routing?

Routing means defining how an application responds to requests made to different URLs.

Q2. What is a route?

A route is a URL path associated with a particular operation or response.

Q3. What are the routes used in your practical?

- / – Home
- /about – About
- /courses – Courses
- /contact – Contact

Q4. What is the purpose of app.listen()?

It starts the server and makes it listen for incoming requests on a specified port.

Q5. What is a port number?

A port number identifies a particular communication endpoint on a computer. Your practical uses port 3000.

Practical 5 – Variables and sum of two numbers

Q1. What is a variable?

A variable is a named container used to store data.

Q2. What are var, let and const?

They are keywords used to declare variables in JavaScript.

- var: Function-scoped.
- let: Block-scoped and can be reassigned.
- const: Block-scoped and cannot be reassigned.

Q3. What is a function?

A function is a reusable block of code designed to perform a particular task.

Q4. What is a parameter?

A parameter is a variable specified in a function definition.

Q5. What is the purpose of return?

It sends a value back from a function to the place where it was called.

Practical 6 – Send JSON response from server

Q1. What is JSON?

JSON stands for JavaScript Object Notation. It is a lightweight format used for storing and exchanging data.

Q2. What is res.json()?

It sends a JSON response to the client.

Q3. What is an array?

An array is a collection of multiple values stored in a single variable.

Q4. What is an object in JavaScript?

An object stores data in key-value pairs.

Example:

```
const student = {
  name: "Amit",
  course: "B.Sc. IT"
};
```

Q5. What is the difference between res.send() and res.json()?

res.send() sends a general response, while res.json() sends a JSON response.

Practical 7 – Apply styling to a React component

Q1. What is a React component?

A component is a reusable part of a React user interface.

Q2. What is CSS?

CSS stands for Cascading Style Sheets. It is used to style web pages.

Q3. What is inline styling?

Inline styling means applying styles directly to an element.

Q4. Why are styles written as JavaScript objects in this practical?

React accepts inline styles as JavaScript objects, where property names use camelCase.

Example: backgroundColor instead of background-color.

Q5. What is React.createElement()?

It creates a React element by specifying the element type, its properties and its children.

Practical 8 – MongoDB database and collection

Q1. What is MongoDB?

MongoDB is a NoSQL, document-oriented database.

Q2. What is a database?

A database is an organised collection of data.

Q3. What is a collection?

A collection is a group of documents in MongoDB.

Q4. What is a document?

A document is a record containing data in key-value pairs.

Q5. What is insertMany()?

It inserts multiple documents into a MongoDB collection.

Q6. What is MongoDB Compass?

MongoDB Compass is a graphical interface used to view and manage MongoDB databases.

Practical 9 – Multiple React components

Q1. Why do we use components in React?

Components make the application reusable, modular and easier to maintain.

Q2. Name the components created in your practical.

- Header
- StudentInfo
- Footer
- App

Q3. What is a functional component?

A functional component is a JavaScript function that returns React elements.

Q4. Can we use one component inside another?

Yes. This is called component composition.

Q5. What is the main component in your practical?

The App component, which renders Header, StudentInfo and Footer.

Practical 10 – Fetch data from API in React

Q1. What is fetch()?

fetch() is a JavaScript function used to make HTTP requests and retrieve data from a server.

Q2. What is useState()?

useState() is a React Hook used to manage state in functional components.

Q3. What is useEffect()?

useEffect() is a React Hook used to perform side effects, such as fetching API data.

Q4. Why is an empty array [] passed to useEffect?

It makes the effect run after the initial render, rather than after every render.

Q5. What is an API endpoint?

It is a specific URL through which an application can access a particular API resource.

Q6. Why do we use response.json()?

To read and convert the response body into JavaScript data.

Practical 11 – Pass data using props

Q1. What are props in React?

Props are inputs used to pass data from a parent component to a child component.

Q2. Are props mutable?

No. Props are read-only inside the receiving component.

Q3. What is a parent component?

A component that renders or uses another component.

Q4. What is a child component?

A component that is rendered inside another component.

Q5. How are props accessed?

Using props.propertyName, for example:

props.name

props.roll

Practical 12 – Add user details using Express and MongoDB

Q1. What is Mongoose?

Mongoose is an ODM (Object Data Modeling) library used to interact with MongoDB through Node.js.

Q2. What is a schema?

A schema defines the structure and rules of documents in a MongoDB collection.

Q3. What is a model?

A model is created from a schema and is used to interact with a MongoDB collection.

Q4. What is POST?

POST is an HTTP method used to send data to a server, commonly to create a new record.

Q5. What is GET?

GET is an HTTP method used to retrieve data from a server.

Q6. What is express.json()?

It is Express middleware that parses incoming JSON request bodies.

Q7. What is Postman?

Postman is a tool used to send and test API requests.

Q8. What is HTTP status code 201?

201 means Created. It indicates that a resource has been successfully created.

Practical 13 – Build a To-do list using React

Q1. What is state in React?

State is data managed by a component that can change over time and cause the component to re-render.

Q2. Which Hook is used in your practical?

useState().

Q3. What is an event in React?

An event is an action such as clicking a button or typing in an input field.

Q4. What is onClick?

It is an event handler that executes a function when an element is clicked.

Q5. What is the purpose of filter()?

It creates a new array containing only the elements that satisfy a given condition.

Q6. What is trim()?

It removes whitespace from the beginning and end of a string.

Practical 14 – Different Express routes

This practical repeats the Express routing topic from Practical 4, with different route responses.

Q1. What is the root route?

The root route is /, which represents the main entry point of a website.

Q2. What does /students return in your code?

It returns a JSON array containing student details.

Q3. What does /status return?

It returns a JSON object containing the server status and success value.

Q4. Can Express have multiple routes?

Yes. We can define multiple routes using methods such as app.get() and app.post().

Practical 15 – React Routing

Q1. What is React Router?

React Router is a library used to implement navigation between pages or views in a React application.

Q2. Which package is installed?

react-router-dom.

Q3. What is BrowserRouter?

It provides routing functionality using the browser's URL history.

Q4. What is the use of Routes?

It groups the route definitions.

Q5. What is the use of Route?

It maps a URL path to a particular component or element.

Q6. What is the use of Link?

It provides navigation between routes without a full browser page reload.

Practical 16 – React frontend with Node and Express backend

Q1. What is full-stack integration?

It means connecting the frontend with the backend so they can exchange data.

Q2. What is CORS?

CORS stands for Cross-Origin Resource Sharing. It controls whether a browser allows requests between different origins.

Q3. Why is cors() used in your practical?

It enables the frontend running on a different origin to access the backend, subject to the server's CORS configuration.

Q4. Why are two terminals required?

One terminal runs the Express backend and the other runs the React frontend.

Q5. Which ports are used?

- Backend: 5000
- Frontend: 5173 (usually)

Q6. How does React receive the backend message?

React uses fetch() to send a request to the Express API and reads the returned JSON response.